Hyung Cho

Forward Deployed Engineer | Full-Stack AI Applications, Integrations, and Problem Solving

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SUMMARY

Builds full-stack AI applications from zero to one: a multi-user RAG app on FastAPI, React, TypeScript, and Supabase with access control enforced in SQL, a support bot that hands off to a human queue, and a native iOS app with an OpenAI-backed coach. Brings 10 years of validation engineering at Zebra Technologies: root-cause analysis across hardware, firmware, and software, and written test plans, procedures, and final reports for 6 shipped products.

EXPERIENCE

Independent AI Engineer | Self-directed projects, Remote **Jun 2025 - Present**

Solo product work, built by directing AI coding agents (Claude Code, Codex).

Purvia, permission-aware RAG and support-bot app, open source

- Built a multi-user RAG app with a FastAPI backend on the raw OpenAI SDK, Supabase auth and storage, multi-format document ingestion, streamed responses, and a React and TypeScript UI, across 114 merged pull requests.
- Built access control into the Postgres vector and keyword retrieval functions so every search runs as the signed-in user, backed by row-level security across 49 SQL migrations.
- Built a support-bot pipeline whose LLM judges fail closed, so any draft that misses a gate escalates to a human operator queue, served through an embeddable widget with rate limits.
- Built a framework-free tool-calling agent loop on the OpenAI API with 4 tools, including web search and text-to-SQL.
- Traced a failing evaluation gate to its own LLM judge rather than the product, then compared five judge designs on replayed drafts and cut wrong verdicts from 38 of 280 to 0 of 300.

Rep Today, native iOS workout app, pre-release

- Built a native iOS workout app in Swift and SwiftUI with Core Data synced through CloudKit, across 214 merged pull requests; it is in internal TestFlight testing.
- Integrated OpenAI through a Cloudflare Worker proxy with App Attest and subscription checks on the server, plus StoreKit 2, Sign in with Apple, Apple Health, and CloudKit on the device.
- Directed development as 214 agent-authored pull requests, 195 with a written risk assessment; later ones separate scenarios validated live in the app from those covered only by tests.

CreativeBridge, K-12 AI story-writing iOS app, internal TestFlight builds

- Migrated the backend from Supabase to Convex with Clerk Google and Apple sign-in and a temporary dual-write, then removed the Supabase paths from the core services.

Volunteer AI/ML Engineer | Omdena, Remote **Apr 2023 - Jun 2025**

Unpaid collaborative ML projects with global volunteer teams.

- Led a DeepLabV3+ cancer-cell segmentation pipeline for histopathology images through training and deployment.
- Worked across time zones with engineers and domain experts, containerizing models in Docker.

Career break and ML retraining **Aug 2018 - Mar 2023**

Self-directed ML study and the FourthBrain ML Engineering bootcamp (2022); 29 public ML projects on GitHub.

Senior Validation Engineer | Zebra Technologies, Agoura Hills, CA **Feb 2008 - Jul 2018**

- Led validation testing for 6 shipped hardware products, writing test plans, procedures, acceptance criteria, and final reports.
- Isolated root causes of failures across hardware, firmware, software, and test infrastructure, and proved each fix.
- Built C# validation tooling that replaced manual test workflows and cut regression test cycles by about 30%.

SKILLS

Full stack: Python, FastAPI, JavaScript, TypeScript, React, SQL, PostgreSQL, Supabase

AI and integrations: OpenAI API, Tool calling, RAG, LLM integration, Cloudflare Workers, StoreKit 2, HealthKit

Delivery and quality: Failure analysis, Test planning, Technical writing, GitHub Actions, Docker, AI coding agents

EDUCATION AND TRAINING

B.S., Electrical Engineering, California Polytechnic State University, San Luis Obispo

FourthBrain ML Engineering bootcamp (2022); DeepLearning.AI deep learning and generative AI courses